

1. How does the shuffling algorithm ensure fairness in randomization?

-> The algorithm iterates through every single slot in the 4×13 deck grid and swaps its card with another card at a randomly generated row and column index. Because `rand()` is seeded with the current time (`srand(time(0))`), every card position has an equal probability of swapping to any location in the grid, ensuring uniform distribution and preventing predictable card patterns.

2. What challenges did you face while implementing poker hand evaluations?

- Rank Frequency Tracking: Determining pairs or sets required counting duplicate ranks across non-sequential array positions. Creating a 13-element frequency counter array simplified tracking how many cards shared the same rank.
- Evaluating Straights: Cards in a hand are not naturally dealt in numerical order. Sorting the hand's rank values first was necessary before checking for consecutive sequences.
- Special Edge Cases: Handling special straight combinations specifically the Ace-low straight (A-2-3-4-5), where Ace counts as 0 rather than 12 which required dedicated conditional checks.